

50 Years of Symbolic Equation Discovery: Revisiting and Improving BACON against Modern Symbolic Regression

Computer Science Tripas Part II Project Proposal

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1: Introduction

Symbolic equation discovery seeks concise, human-readable formulas that generalise beyond observed data. Modern practice uses symbolic regression, a machine learning subfield that infers symbolic mathematical expressions from data, to search expression trees [1]. It is powerful but on harder problems can be compute-demanding, sensitive to settings, and less transparent than invariant-first methods [2].

This project revisits the BACON approach: detect simple regularities, layer derived variables, and produce transparent laws [3, 4]. I will implement BACON.3 and a separate improved variant inspired by recent reconstructions, then compare both to PySR (widely used open-source baseline) under identical settings [5]. If time allows, I will also implement a small symbolic regression solver.

2: Work to be done

This project implements and studies transparent, invariant-first equation discovery. Namely, the project shall:

- Implement BACON.3 from first principles.
- Build an improved BACON variant, as per Miller’s upgrades [6].
- Evaluate fairly under shared settings, using PySR as a reference.
- If time allows, the extensions as per below.

2.1 BACON.3

BACON.3 discovers equations by layered composition of simple invariants [7]. At each layer it generates a small set of primitive composites from the current variable pool (products, ratios, small integer powers) and probes two cues [3, 4, 7]:

Near constancy the composite varies little across the table and suggests a proportionality or conserved quantity.

Monotonicity the composite changes consistently with the target and suggests a direct functional link.

Composites that pass are promoted to derived variables and returned to the pool for the next layer. The process stops when no candidate meets these cues or a shallow depth cap is reached. A closing relation is then written among the accepted variables, yielding a concise formula with an auditable trail of decisions.

Success Criteria 1: a working BACON.3 with readable decision logs that rediscover several textbook relations under small-data noise.

2.2 Improving BACON.3

A baseline BACON.3 approach can stall on small, noisy datasets: strict consensus blocks progress, noise propagates across layers, and fixed thresholds are brittle. I aim to improve it with a selective, noise-tolerant, and reproducible pipeline that preserves interpretability.

Following Miller’s “BACON.7”, I will improve BACON.3 by [6]:

- Implementing a layer-method (e.g., popular, Min-MSE) to choose one expression per layer.
- Applying it by symmetry (reuse the same algebraic pattern on every equivalent subset of data).
- Taking the numerical average within each subset after transformation.
- Adding Δ branch pruning (stop a branch when layer values vary less than a small tolerance around their mean).
- Using a managed pairwise law finder with an iteration cap and auto-relaxed tolerances.

Success Criteria 2: implement this improved BACON and scientifically evaluate it against the BACON.3 baseline on a standard benchmark set, reporting results and decision logs regardless of outcome.

2.3 Evaluation

All methods will use the same settings, and I will build a small evaluation framework for core checks such as exact recovery, extrapolation, stability, and efficiency.

The test set will include:

- clean synthetics (e.g., $y = x_1x_2$, $y = x_1/x_2$);
- well-known textbook laws (e.g., Ohm’s law, $V = IR$);
- realistic public datasets (e.g., AI Feynman repository [8]).

Results will summarise, e.g., exactness, extrapolation beyond training, stability under small resamples, and time/proposals to good formulas, with additional sensible checks from recent literature where appropriate.

Success Criteria 3: runs are reproducible from a fresh checkout; methods use shared settings; key behaviours are reported clearly on the chosen tasks.

2.4 Extension: Mini Symbolic Regression Model

After the core study, I will add a small SR solver to contrast direct expression search with the invariant-first pipeline under the same settings. It will use a small population, evolutionary operators (mutation and crossover), optional constant refits, a parsimony preference, and clear logs; if time allows, one additional variant to contrast search behaviour. The extension will run on a small subset.

Success Criteria 4: a runnable mini SR with clear logs; if time allows, one additional variant.

2.5 Other Potential Extensions

Further variants of improved BACON.3, e.g., alternative evidence aggregation, different parsimony schemes, or light search guidance.

2.6 Risks & Mitigations

If PySR is difficult to install or unstable on my laptop, I will substitute my mini SR under identical settings and document the change. If realistic public data proves messy, I will fall back to scripted noise injections on textbook laws and document ranges/noise.

3: Starting Point

No prior code. No prior experience with BACON/PySR; clean-room implementations only. Python proficiency and standard scientific-Python stack; familiar with testing and repo hygiene.

4: Resource Declaration

Primary machine: 14-inch MacBook Pro (2024, M4, 24 GB RAM, 1 TB). Notes local + cloud backup; dissertation in L^AT_EX. Code on GitHub with version control.

Software: Python 3 with NumPy/SciPy/Pandas, SymPy, Matplotlib; PySR from its public repo.

5: Timeline

5.1 Michaelmas Term

Weeks 1–2 (Sat 11 Oct – Fri 24 Oct)

- Read core literature.
- Compile a methods dossier.
- Set up repo/env/scaffold.
- **Milestone:** PDF dossier sent to D2D Supervisor; scaffold builds.

Weeks 3–4 (Sat 25 Oct – Fri 7 Nov)

- Implement BACON.3.
- Write unit tests.
- Achieve first recoveries.
- **Milestone:** BACON.3 reproduces toy laws with logs.
- **Commitments:** NLP Assignment 1 [Thu 30 Oct].

Weeks 5–6 (Sat 8 Nov – Fri 21 Nov)

- Prepare initial datasets.
- **Milestone:** Dataset scripts and ready-to-run catalogue.
- **Commitments:** NLP Assignment 2 [Thu 13 Nov].

Weeks 7–8 (Sat 22 Nov – Fri 5 Dec)

- Run first BACON.3 sweep; make quick plots.
- Draft improvement plan.
- **Milestone:** Baseline results + one-page issues/upgrade memo.
- **Commitments:** NLP Assignment 3 [Thu 4 Dec].

5.2 Michaelmas Vacation

Weeks 1–2 (Sat 6 Dec – Fri 19 Dec)

- Finish any term spillover.
- Finalise improved BACON design; write pseudocode and interfaces.
- **Milestone:** Signed design; open issues closed or listed.

Weeks 3–4 (Sat 20 Dec – Fri 2 Jan)

- Implement improved BACON.
- Build evaluation framework and regenerate sample figures.
- **Milestone:** Improved BACON runs; framework produces figures.

Weeks 5–6 (Sat 3 Jan – Fri 16 Jan)

- Add PySR and run full core sweep.
- Collect logs and plots.
- **Milestone:** Core comparison dataset ready for analysis.

Week 7 (Sat 17 Jan – Fri 23 Jan)

- Tidy up and rerun stragglers.
- **Milestone:** All prior milestones complete.

5.3 Lent Term

Weeks 1–2 (Sat 24 Jan – Fri 6 Feb)

- Analyse the core sweep.
- Produce clean plots and two short case studies.
- Prepare progress report and presentation.
- **Milestone:** Core analysis pack delivered; Progress Report submitted (Fri 6 Feb).

Weeks 3–4 (Sat 7 Feb – Fri 20 Feb)

- Implement the mini SR prototype.
- Run on a small subset under shared settings; first comparison with BACON.
- **Milestone:** Mini SR baseline running with initial results.

Weeks 5–6 (Sat 21 Feb – Fri 6 Mar)

- Finish mini-SR baseline runs and any BACON reruns.
- Tidy plots.
- If time allows: add a simple SR variant and run it on the small subset.
- **Milestone:** Runs to date complete with a brief comparison note.
- **Commitments:** DNN Assignment 1.

Weeks 7–8 (Sat 7 Mar – Fri 20 Mar)

- Consolidate results; final figure shortlist.
- Buffer & fixes: reruns, log cleanup, prepare appendix materials.
- **Milestone:** Results locked; artefacts ready for write-up.
- **Commitments:** DNN Assignment 2.

5.4 Lent Vacation

Weeks 1–2 (Sat 21 Mar – Fri 3 Apr)

- Revise; draft *Introduction* and *Preparation*.
- **Milestone:** Draft sent to supervisor/DoS.

Weeks 3–4 (Sat 4 Apr – Fri 17 Apr)

- Revise; draft *Implementation*.
- **Milestone:** Draft sent.

Weeks 5–6 (Sat 18 Apr – Fri 1 May)

- Revise; draft *Evaluation* and *Conclusion*.
- **Milestone:** Full draft sent.

5.5 Easter Term

Weeks 1–2 (Sat 2 May – Fri 15 May)

- Finish any evaluation; edit dissertation; tidy repo; revise.
- **Milestone:** Submit dissertation and source code [Fri 15 May].

Bibliography

- [1] N. Makke and S. Chawla. Interpretable scientific discovery with symbolic regression: A review. *Artificial Intelligence Review*, 57, Article 10622, 2024. <https://doi.org/10.1007/s10462-023-10622-0>.
- [2] G. Aldeia *et al.* Call for Action: towards the next generation of symbolic regression benchmark. *arXiv:2505.03977*, 2025. <https://arxiv.org/abs/2505.03977>.
- [3] P. Langley. BACON: A production system that discovers empirical laws. In *Proceedings of IJCAI-77*, pp. 344–350, 1977.
- [4] P. Langley. Data-Driven Discovery of Physical Laws. *Cognitive Science*, 5:31–54, 1981.
- [5] M. Cranmer. Interpretable Machine Learning for Science with PySR and SymbolicRegression.jl. *arXiv:2305.01582*, 2023. <https://arxiv.org/abs/2305.01582>.
- [6] J. Miller and S. Banerjee. The BACON system for equation discovery from scientific data: Transforming classical AI with modern ML approaches. *AAAI Fall Symposium Series: Integrated Approaches to Computational Scientific Discovery*, 2024.
- [7] P. Langley. Rediscovering physics with Bacon.3. In *Proceedings of IJCAI-79*, pp. 505–507, 1979.
- [8] S. M. Udrescu and M. Tegmark. AI Feynman: A physics-inspired method for symbolic regression. *Science Advances*, 6(16):eaay2631, 2020.